Building Connections Between Topological Spaces With Games

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Introduction

Main Themes:

- Building new topological spaces out of old ones.
- Connecting clustering properties to compactness properties, or different kinds of compactness properties to each other.
- Using games and strategies to capture topological properties.
- Using topological connections to create strategic connections.

New Spaces From Old Ones

For the duration of this talk, when I say **derived space**, I mean one of the following:

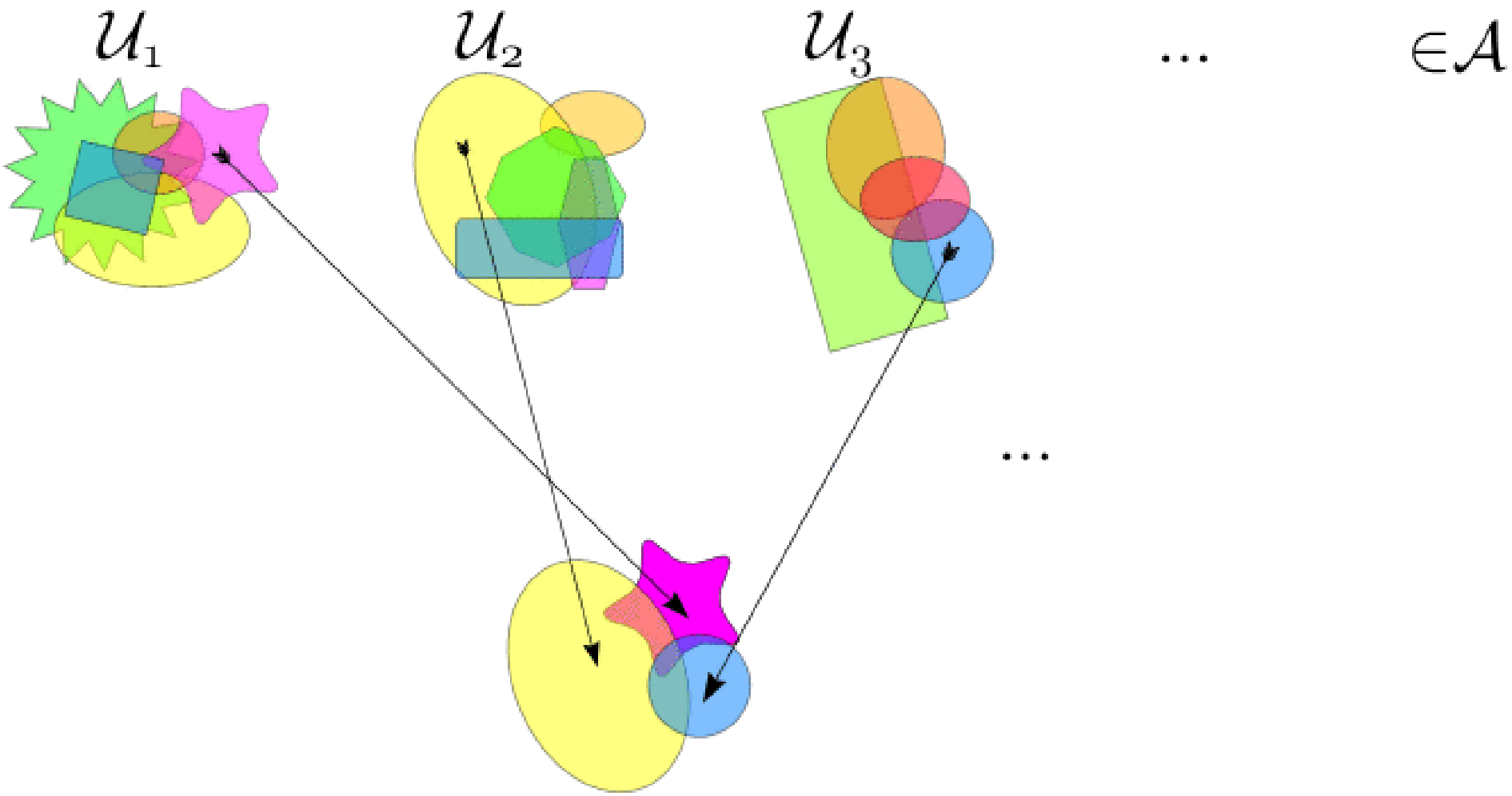
- The continuous real-valued functions with domain X. This could have the topology of pointwise convergence or uniform convergence on compact sets.
- Upper semi-continuous set valued (USCO) maps with domain X.
- The hyperspace of closed/compact subsets of X. This could have the Vietoris topology, the Fell topology, or the Pixley-Roy topology.

Variations on Compactness

Recall that a space X is **compact** if every open cover of X has a finite subcover. We say

- X is **Lindelöf** if every open cover of X has a *countable* subcover.
- X is **Rothberger** if for every sequence $\langle \mathcal{U}_n \rangle$ of open covers of X, there are open sets $U_n \in \mathcal{U}_n$ so that $X = \bigcup_n U_n$.
- X is **Menger** if for every sequence $\langle \mathcal{U}_n \rangle$ of open covers of X, there are finite $F_n \subseteq \mathcal{U}_n$ so that $X = \bigcup_n F_n$.

Rothberger

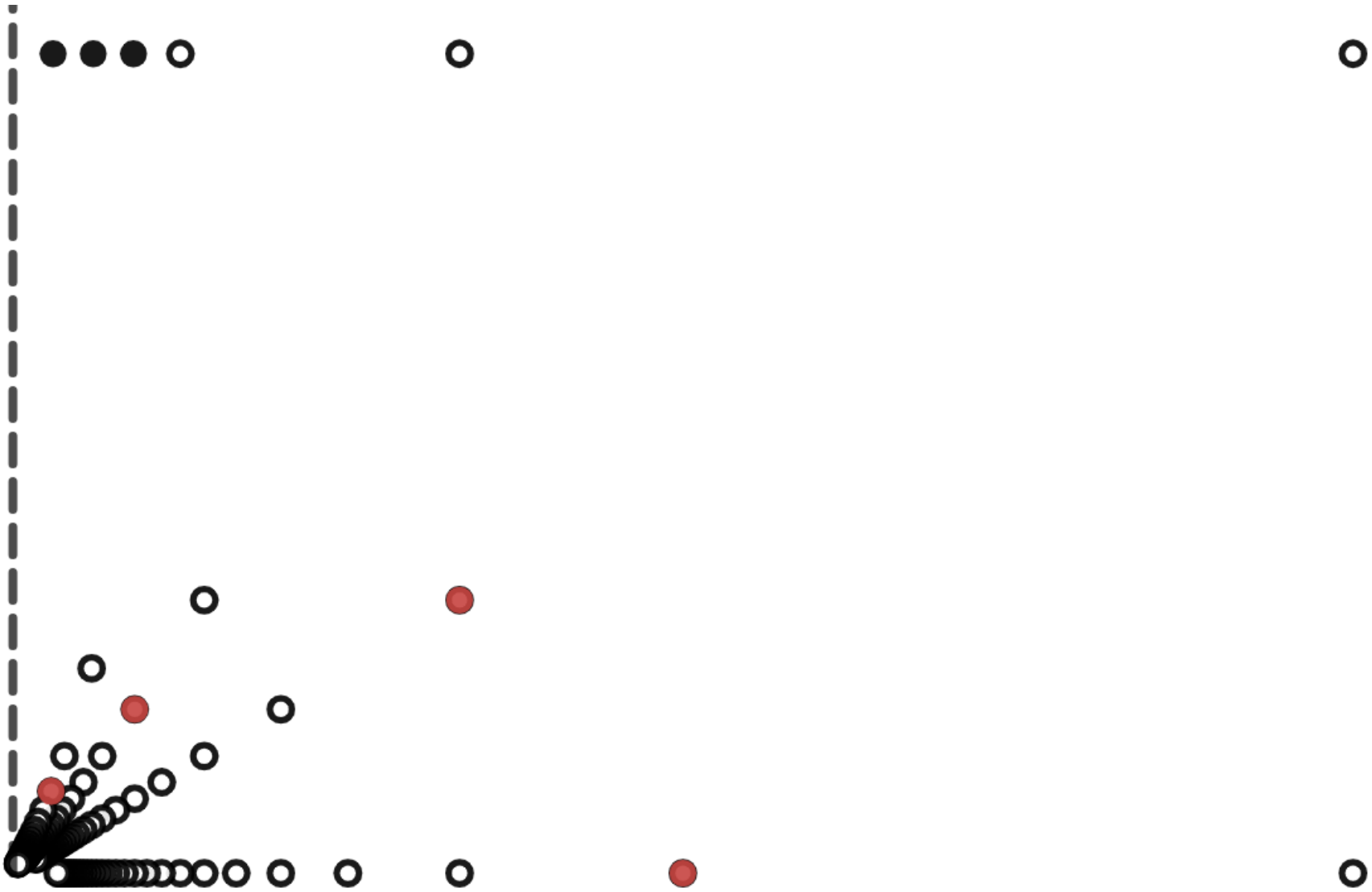


Clustering

Recall that a sequence $\langle x_n \rangle$ clusters to x if for every neighborhood U of x and every N, there is a $k \geq N$ so that $x_k \in U$. Alternately, $x \in \overline{\{x_n : n \in \mathbb{N}\}}$.

- X has **countable tightness** if whenever $A \subseteq X$ and $x \in \overline{A}$, there is a countable $B \subseteq A$ so that $x \in \overline{B}$.
- X has **strong countable fan-tightness at** x if whenever $A_n \subseteq X$ and $x \in \overline{A_n}$ for all n, there are $x_n \in A_n$ so that $x \in \overline{\{x_n : n \in \mathbb{N}\}}$
- X has **countable fan-tightness at** x if whenever $A_n \subseteq X$ and $x \in \overline{A_n}$ for all n, there are finite $F_n \subseteq A_n$ so that $x \in \overline{\bigcup_n F_n}$.

Strong Countable Fan Tightness



Selection Principles (Menger, Hurewicz 1924) (Scheepers 1996)

Suppose that $\mathcal{A}$ and $\mathcal{B}$ are collections (typically of topologically interesting objects).

$S_1(\mathcal{A}, \mathcal{B})$

$S_1(\mathcal{A}, \mathcal{B})$ means that for all sequences $\langle A_n \rangle$ consisting of elements of $\mathcal{A}$, there are choices $x_n \in A_n$ so that $\{x_n : n \in \mathbb{N}\} \in \mathcal{B}$.

$S_{\text{fin}}(\mathcal{A}, \mathcal{B})$

$S_{\text{fin}}(\mathcal{A}, \mathcal{B})$ means that for all sequences $\langle A_n \rangle$ consisting of elements of $\mathcal{A}$, there are finite $F_n \subseteq A_n$ so that $\bigcup_n F_n \in \mathcal{B}$.

Notation

- Let $\mathcal{O}(X)$ denote the open covers of X.
- Let $\Omega_{X,x}$ denote the sets $A \subseteq X$ such that $x \in \overline{A}$.

Note that

- X is Rothberger if and only if $S_1(\mathcal{O}(X), \mathcal{O}(X))$.
- X is Menger if and only if $S_{\text{fin}}(\mathcal{O}(X), \mathcal{O}(X))$.
- X has strong countable fan tightness at x if and only if $S_1(\Omega_{X,x}, \Omega_{X,x})$.
- X has countable fan tightness at x if and only if $S_{\text{fin}}(\Omega_{X,x}, \Omega_{X,x})$.

Strategies

- A **perfect information (PI) strategy** for either player One or Two is a strategy for responding to the other player that takes as inputs all of the previous plays of the game.
- A **Markov strategy** for player Two is a strategy that takes as inputs the current turn number and the most recent play of player One.
- A **pre-determined (PD) strategy** for player One is a strategy where the only input is the current turn number.
- A strategy is **winning** if following the strategy guarantees that the player will win the game.

Strategies

- Playing according to a PI strategy for One:

I	$\sigma(\emptyset)$	$\sigma(x_0)$	$\sigma(x_0, x_1)$	$\cdots$
II	x_0	x_1	x_2	$\cdots$

- Playing according to a PI strategy for Two:

I	A_0	A_1	A_2	$\cdots$
II	$\tau(A_0)$	$\tau(A_0, A_1)$	$\tau(A_0, A_1, A_2)$	$\cdots$

- Playing according to a Markov strategy for Two:

I	A_0	A_1	A_2	$\cdots$
II	$\tau(A_0, 0)$	$\tau(A_1, 1)$	$\tau(A_2, 2)$	$\cdots$

- Playing according to a PD strategy for One:

I	$\sigma(0)$	$\sigma(1)$	$\sigma(2)$	$\cdots$
II	x_0	x_1	x_2	$\cdots$

A Strength Hierarchy

For the selection game $G_1(\mathcal{A}, \mathcal{B})$:

$$\begin{array}{c}
\text{Two has a winning Markov Strategy} \\
\Downarrow \\
\text{Two has a winning PI strategy} \\
\Downarrow \\
\text{One has no winning PI strategy} \\
\Downarrow \\
\text{One has no winning PD strategy} \iff S_1(\mathcal{A}, \mathcal{B})
\end{array}$$

A Return to Compactness

Note the following.

- X is Rothberger iff player one does not have a winning pre-determined strategy in $G_1(\mathcal{O}(X), \mathcal{O}(X))$.
- X is Menger iff player one does not have a winning pre-determined strategy in $G_{\text{fin}}(\mathcal{O}(X), \mathcal{O}(X))$.
- X is Lindelöf iff player one does not have a winning constant strategy in $G_1(\mathcal{O}(X), \mathcal{O}(X))$.

A Return to Clustering

- X has strong countable fan tightness if for all $x \in X$, player one does not have a winning pre-determined strategy in $G_1(\Omega_{X,x}, \Omega_{X,x})$.
- X has countable fan tightness if for all $x \in X$, player one does not have a winning pre-determined strategy in $G_{\text{fin}}(\Omega_{X,x}, \Omega_{X,x})$.

Game Equivalence

Definition

Define $G_1(\mathcal{A},\mathcal{C}) \leq_{\mathrm{II}} G_1(\mathcal{B},\mathcal{D})$ as the conjunction of the following implications.

1. Two has Mark in $G_1(\mathcal{A},\mathcal{C}) \implies$ Two has a Mark in $G_1(\mathcal{B},\mathcal{D})$
2. Two has PI strat in $G_1(\mathcal{A},\mathcal{C}) \implies$ Two has a PI strat in $G_1(\mathcal{B},\mathcal{D})$
3. One has no PI strat in $G_1(\mathcal{A},\mathcal{C}) \implies$ One has no PI strat in $G_1(\mathcal{B},\mathcal{D})$
4. One has no PD strat in $G_1(\mathcal{A},\mathcal{C}) \implies$ One has no PD strat in $G_1(\mathcal{B},\mathcal{D})$

- This relation is transitive.
- if $G_1(\mathcal{A},\mathcal{C}) \leq_{\mathrm{II}} G_1(\mathcal{B},\mathcal{D})$ and $G_1(\mathcal{B},\mathcal{D}) \leq_{\mathrm{II}} G_1(\mathcal{A},\mathcal{C})$, we say the two games are **equivalent** and write $G_1(\mathcal{A},\mathcal{C}) \equiv G_1(\mathcal{B},\mathcal{D})$.
- There is a G_{fin} version of all of this.

Scheeper's Monotonicity Law

Let $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$ be collections.

Monotonicity

- If $\mathcal{B} \subseteq \mathcal{A}$, then $G_{\square}(\mathcal{A}, \mathcal{C}) \leq_{\mathrm{II}} G_{\square}(\mathcal{B}, \mathcal{C})$.
- If $\mathcal{D} \subseteq \mathcal{C}$, then $G_{\square}(\mathcal{A}, \mathcal{D}) \leq_{\mathrm{II}} G_{\square}(\mathcal{A}, \mathcal{C})$.

Another Variation on Clustering

Countable Dense Fan Tightness

X has strong countable dense fan tightness at x if whenever $\langle D_n \rangle$ is a sequence of dense subsets of X, there are points $x_n \in D_n$ so that $x \in \overline{\{x_n : n \in \mathbb{N}\}}$.
If we let $\mathcal{D}_X$ represent the dense subsets of X, this property is $S_1(\mathcal{D}_X, \Omega_{X,x})$

As an application of Scheeper's Monotonicity law,
$G_1(\Omega_{X,x}, \Omega_{X,x}) \leq_{\mathrm{II}} G_1(\mathcal{D}_X, \Omega_{X,x})$ for any space X and point $x \in X$.

The Fell Topology on the Closed Subsets of X

Let $\mathbb{F}(X)$ denote the collection of closed subsets of X.

The Upper Fell Topology (Fell 1962)

The **Upper Fell Topology** on $\mathbb{F}(X)$ is generated by basic open sets of the form

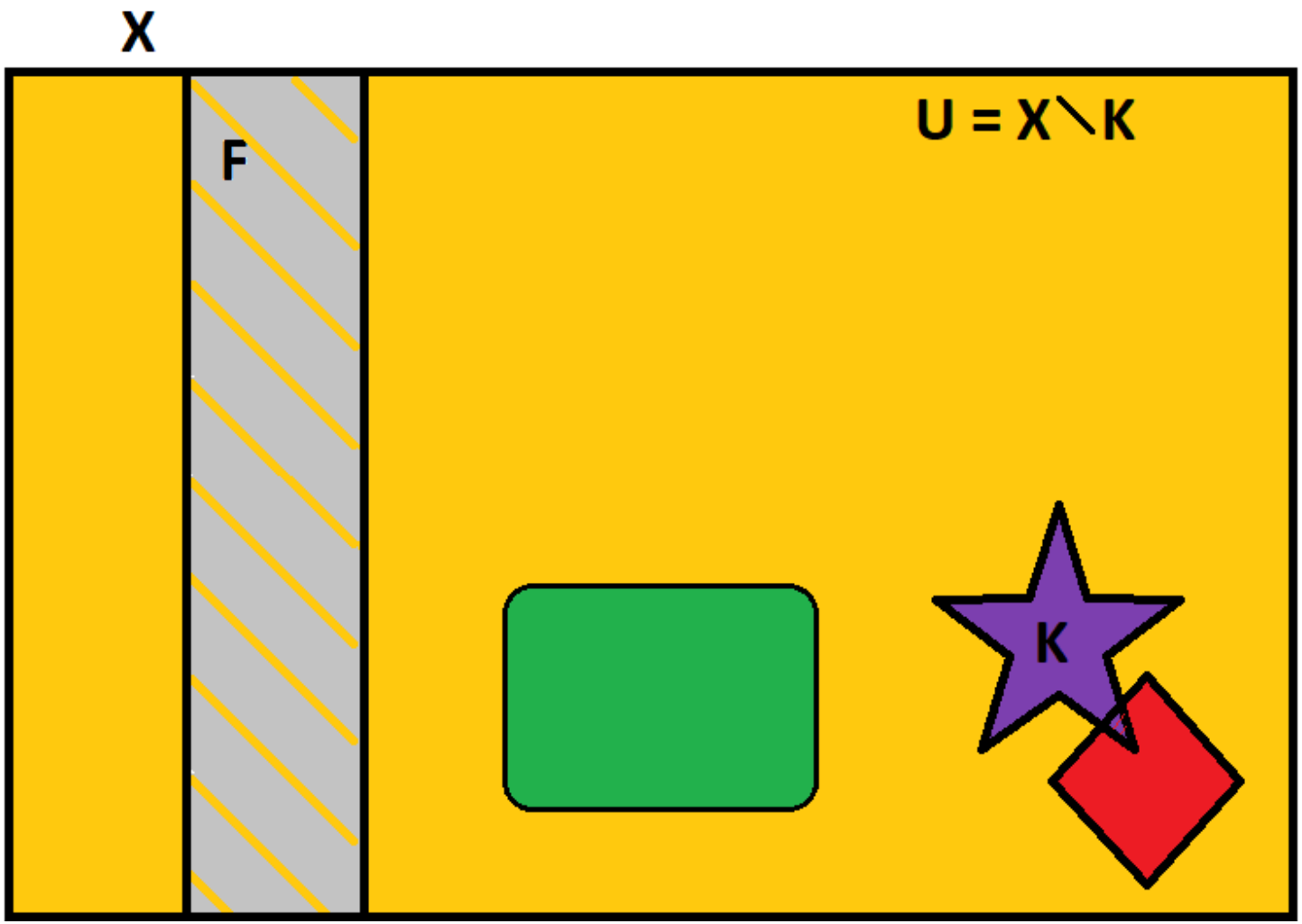
$$(U)^+ := \{F \in \mathbb{F}(X) : F \subseteq U\}$$

where U is the complement of a compact set.

A basic neighborhood of F has the form $(X \setminus K)^+$ where K is compact and $F \cap K = \emptyset$.
Let $\mathbb{F}^+(X)$ be the closed subsets of X with the upper Fell topology.

The Fell Topology on the Closed Subsets of X

F with an open neighborhood $(X \setminus K)^+$.



The Fell Topology on the Closed Subsets of X

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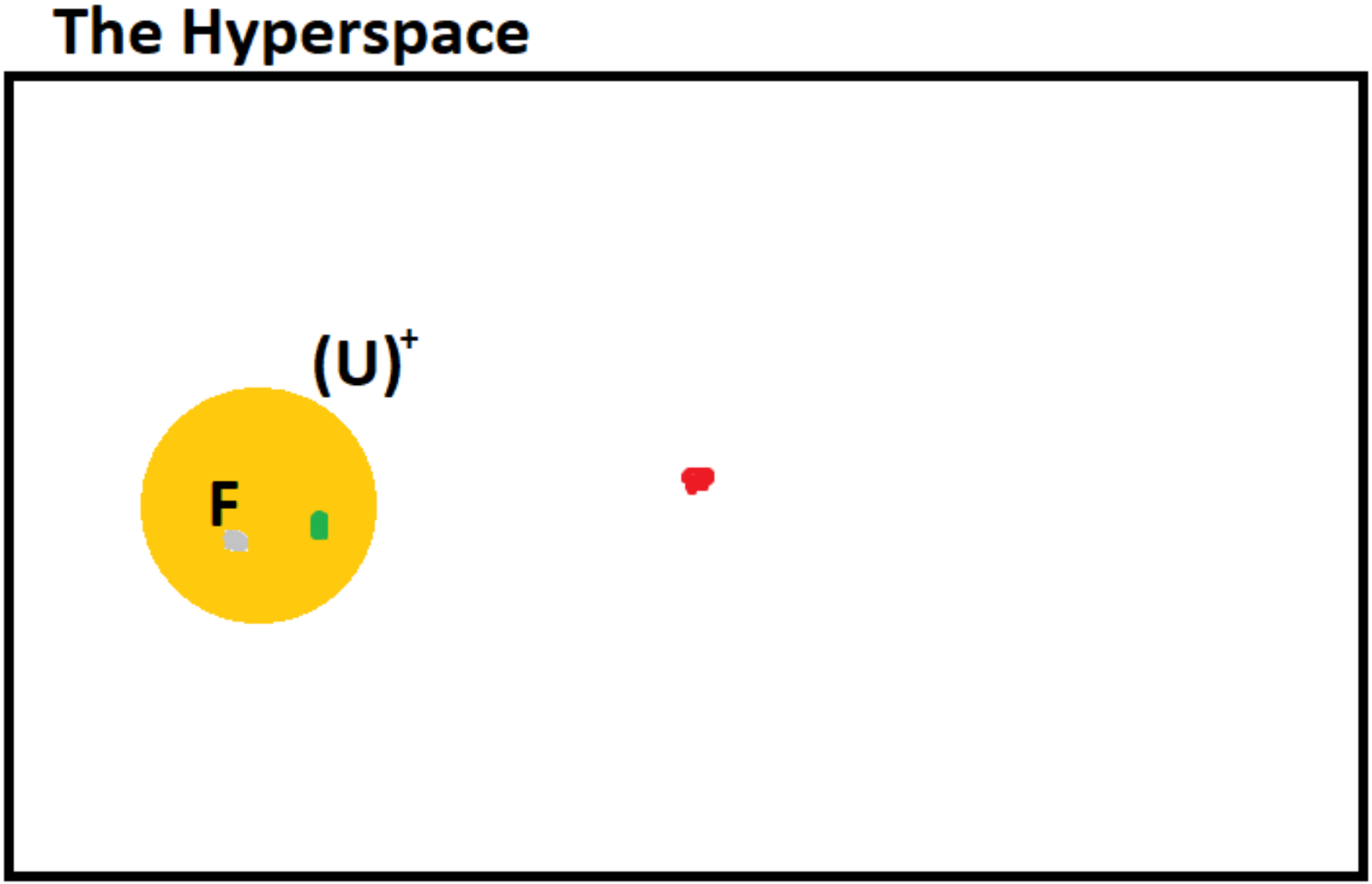


A Topological Connection

- $U \subseteq X$ is open if and only if $X \setminus U$ is a point in $\mathbb{F}^+(X)$.
- If $D \subseteq \mathbb{F}^+(X)$ is dense, then $\{X \setminus F : F \in D\}$ is an open cover of X.
- In fact, if $K \subseteq X$ is compact, then there is an $F \in D$ so that $K \subseteq X \setminus F$.

k-Covers

A non-trivial open cover $\mathscr{U}$ of X (meaning $X \notin \mathscr{U}$) is a **k-cover** if for all compact $K \subseteq X$, there is a $U \in \mathscr{U}$ so that $K \subseteq U$.
$\mathcal{K}(X)$ is the collection of k-covers of X.

Proposition

- $\mathscr{U} \in \mathcal{K}(X)$ if and only if $\{X \setminus U : U \in \mathscr{U}\} \in \mathscr{D}_{\mathbb{F}^+(X)}$.
- $D \in \mathscr{D}_{\mathbb{F}^+(X)}$ if and only if $\{X \setminus F : F \in D\} \in \mathcal{K}(X)$.

Bijective Translation

Caruvana, Holshouser 2021

Assume that $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$ are collections and that $\bigcup \mathcal{C} \subseteq \bigcup \mathcal{A}$ and $\bigcup \mathcal{D} \subseteq \bigcup \mathcal{B}$. Suppose that there is a bijection $\beta : \bigcup \mathcal{A} \to \bigcup \mathcal{B}$ so that

- $A \in \mathcal{A}$ if and only if $\beta[A] \in \mathcal{B}$, and
- $C \in \mathcal{C}$ if and only if $\beta[C] \in \mathcal{D}$.

Then $G_{\square}(\mathcal{A}, \mathcal{C}) \equiv G_{\square}(\mathcal{B}, \mathcal{D})$.

Using Bijective Translation

Caruvana, Holshouser 2021

$G_\square(\mathcal{K}(X), \mathcal{K}(X)) \equiv G_\square(\mathcal{D}_{\mathbb{F}^+(X)}, \mathcal{D}_{\mathbb{F}^+(X)})$.

We can prove this by defining $\beta : \mathcal{T}_X \to \mathbb{F}^+(X)$ as $\beta(U) = X \setminus U$ and applying the proposition.

Using Bijective Translation

Caruvana, Holshouser 2021

$G_\square(\mathcal{K}(X), \mathcal{K}(X)) \equiv G_\square(\mathcal{D}_{\mathbb{F}^+(X)}, \mathcal{D}_{\mathbb{F}^+(X)})$.

We can prove this by defining $\beta : \mathcal{T}_X \to \mathbb{F}^+(X)$ as $\beta(U) = X \setminus U$ and applying the proposition.

Brought down to the selection principle level, this implies that X is k-Rothberger if and only if $\mathbb{F}^+(X)$ is something called selectively separable.

Full Vietoris

The **Vietoris Topology** on $\mathbb{F}(X)$ is generated by basic open sets of the form

$$[U; V_1, \cdots, V_n] = \{F \in \mathbb{F}(X) : F \subseteq U \text{ and } F \cap V_j \neq \emptyset \text{ for } 1 \leq j \leq n\}$$

where $U, V_1, \cdots, V_n \subseteq X$ are open. Let $\mathbb{F}(X)$ denote the hyperspace of closed sets with the Vietoris topology.

In 2016, Li introduced the concept of c_v-covers and showed that $S_1(c_v(X), c_v(X))$ is equivalent to $S_1(\mathcal{D}_{\mathbb{F}(X)}, \mathcal{D}_{\mathbb{F}(X)})$ (among other results).

Caruvana, Holshouser 2021

$G_\square(c_v(X), c_v(X)) \equiv G_\square(\mathcal{D}_{\mathbb{F}(X)}, \mathcal{D}_{\mathbb{F}(X)})$

Pixley-Roy

- The Pixley-Roy hyperspace was introduced by Pixley and Roy in 1969 as a way of generating interesting counter examples.
- In 2014, Sakai studied star versions of the Menger property and established a connection between this star Menger property on the Pixley-Roy hyperspace and the Menger game on the ground space.
- In 2022, we extended this to all strategy levels.

The Space of Continuous Functions

Set $C(X)$ to be collection of all continuous functions $f : X \to \mathbb{R}$.

The Topology of Point-Wise Convergence

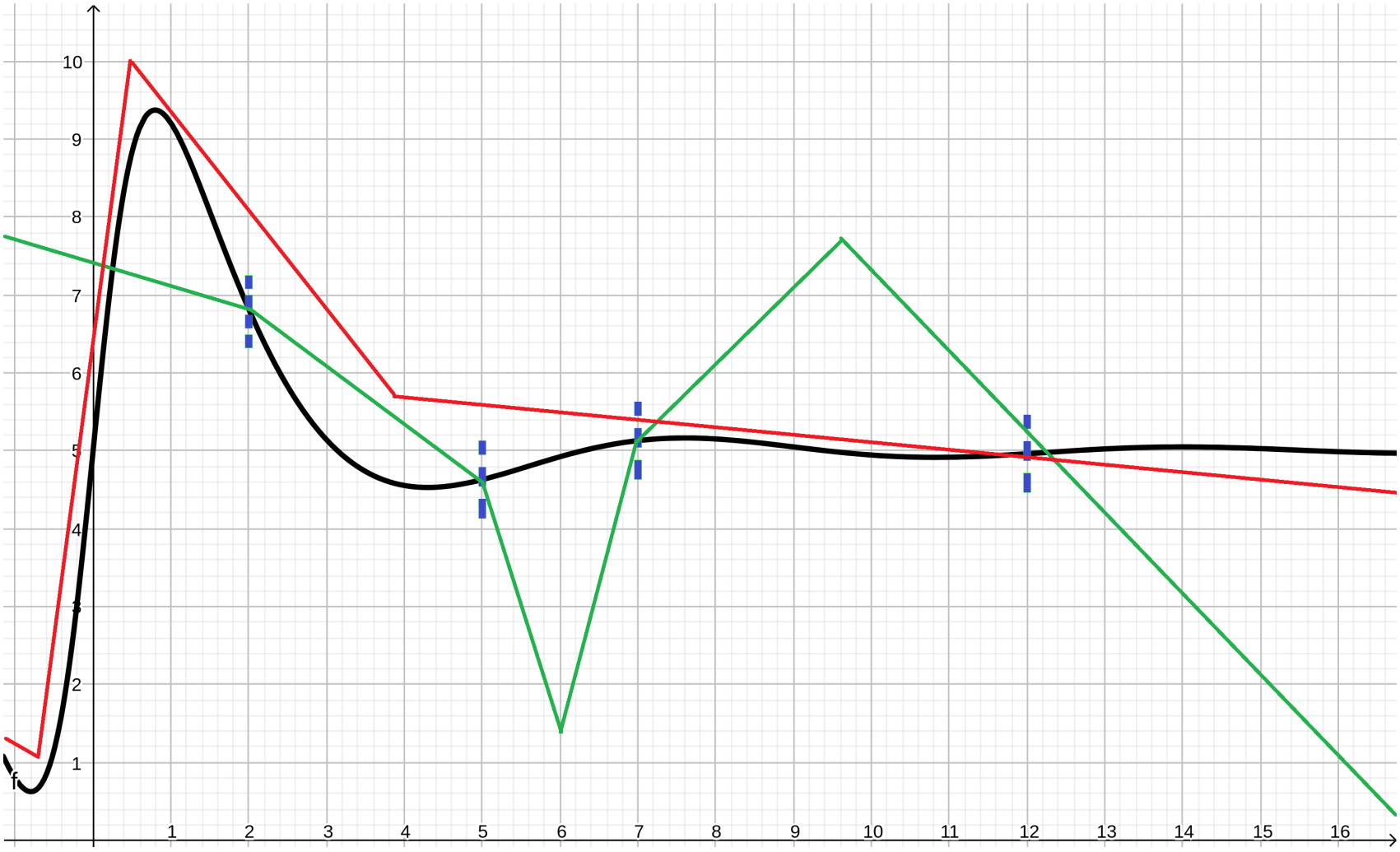
$C(X)$ with this topology will be denoted $C_p(X)$. The open sets are generated by sets of the form:

$$[f; \{x_0, \cdots, x_n\}, \varepsilon] = \{g : |f(x_0) - g(x_0)| < \varepsilon, \cdots, |f(x_n) - g(x_n)| < \varepsilon\}$$

where f is continuous, $x_0, \cdots, x_n \in X$, and $\varepsilon > 0$.

Common Topologies on the Space of Continuous Functions

f with a neighborhood $[f; F, \varepsilon]$.



A Topological Connection

- If $f : X \to \mathbb{R}$ is continuous, and $n \in \mathbb{N}$, then $f^{-1}[(-2^{-n}, 2^{-n})]$ is open.
- If $A \subseteq C_p(X)$ has $\mathbf{0}$ in its closure, then for a fixed n,

$$\mathscr{U} = \{f^{-1}[(-2^{-n}, 2^{-n})] : f \in A\}$$

  is an open cover of X.
- In fact, if $F \subseteq X$ is finite, then there is a $U \in \mathscr{U}$ so that $F \subseteq U$.

ω-Covers

A non-trivial open cover $\mathscr{U}$ of X is an **ω-cover** if for all finite $F \subseteq X$, there is a $U \in \mathscr{U}$ so that $F \subseteq U$.
$\Omega(X)$ is the collection of ω-covers of X.

Proposition

If $A \in \Omega_{C_p(X),\mathbf{0}}$, then

$$\mathscr{U}(A, n) := \{f^{-1}[(-2^{-n}, 2^{-n})] : f \in A\}$$

is an ω-cover. I.e. $\mathscr{U}(A, n) \in \Omega(X)$.

Proof.

Suppose $A \in \Omega_{C_p(X),\mathbf{0}}$. Let $F \subseteq X$ be finite. Then $[\mathbf{0}; F, 2^{-n}]$ is an open nhood of $\mathbf{0}$. So there is an $f \in A$ so that $f \in [\mathbf{0}; F, 2^{-n}]$. Then $F \subseteq f^{-1}[(-2^{-n}, 2^{-n})]$.

$\square$

ω-Covers

Proposition

Suppose $f_n \in C_p(X)$ are so that

$$\{f_n^{-1}[(-2^{-n}, 2^{-n})]\} \in \Omega(X).$$

Then $\{f_n : n \in \mathbb{N}\} \in \Omega_{C_p(X),\mathbf{0}}$.

Proof.

Consider a basic open nhood $[\mathbf{0}; F, \varepsilon]$. Since F is finite, there is an n so that $2^{-n} < \varepsilon$ and $F \subseteq f_n^{-1}[(-2^{-n}, 2^{-n})]$. Thus $f_n \in [\mathbf{0}; F, \varepsilon]$. $\square$

Turn Based Translation

Caruvana, Holshouser 2020

Assume that $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$ are collections and that $\bigcup \mathcal{C} \subseteq \bigcup \mathcal{A}$ and $\bigcup \mathcal{D} \subseteq \bigcup \mathcal{B}$. Suppose that there is a function $\varphi : \bigcup \mathcal{B} \times \mathbb{N} \to \bigcup \mathcal{A}$ so that

- If $B \in \mathcal{B}$ and $n \in \omega$, then $\varphi[B \times \{n\}] \in \mathcal{A}$, and
- If $y_n \in B_n$ and $\{\varphi(y_n, n) : n \in \mathbb{N}\} \in \mathcal{C}$, then $\{y_n : n \in \mathbb{N}\} \in \mathcal{D}$.

Then $G_1(\mathcal{A}, \mathcal{C}) \leq_{\mathrm{II}} G_1(\mathcal{B}, \mathcal{D})$.

Turn Based Translation

$G_1(\mathcal{A}, \mathcal{C})$: "$\varphi$" $G_1(\mathcal{B}, \mathcal{D})$:

I	$\varphi[B_0 \times \{0\}]$	$\varphi[B_1 \times \{1\}]$	$\cdots$
II	$\varphi(y_0, 0)$	$\varphi(y_1, 1)$	$\cdots$

I	B_0	B_1	$\cdots$
II	y_0	y_1	$\cdots$

"φ^{-1}"

Two can win implies Two can win

① $B \in \mathcal{B}$ and $n \in \omega$ $\Rightarrow$ $\varphi[B \times \{n\}] \in \mathcal{A}$

② $y_n \in B_n$ and $\{\varphi(y_n, n) : n \in \omega\} \in \mathcal{C}$ $\Rightarrow$ $\{y_n : n \in \omega\} \in \mathcal{D}$

Using Turn Based Translation

Clontz, Holshouser 2019

$G_1(\Omega(X), \Omega(X)) \leq_{\mathrm{II}} G_1(\Omega_{C_p(X),\mathbf{0}}, \Omega_{C_p(X),\mathbf{0}})$.

We can prove this by defining $\varphi : C_p(X) \times \mathbb{N} \to \mathcal{T}_X$ as $\varphi(f, n) = f^{-1}[(-2^{-n}, 2^{-n})]$ and applying our facts about ω-covers.

Complete Regularity

Definition

Recall that space X is **completely regular** if for every point $x \in X$ and closed set $F \subseteq X$ with $x \notin F$, there is a continuous function $f : X \to [0, 1]$ so that $f(x) = 0$ and $f|_F = \mathbf{1}$.

Note that you can also find continuous functions to separate finite sets from closed sets, and even separate compact sets from closed sets.

Complete Regularity

Suppose that X is Hausdorff and completely regular.

Proposition

If $\mathscr{U} \in \Omega(X)$, then

$$A(\mathscr{U}) := \{f : (\exists U \in \mathscr{U})[f|_{X \setminus U} = \mathbf{1}]\} \in \Omega_{C_p(X),\mathbf{0}}.$$

Proof.

Suppose $\mathscr{U} \in \Omega(X)$. Consider a basic open nhood $[\mathbf{0}; F, \varepsilon]$. There is a $U \in \mathscr{U}$ so that $F \subseteq U$. There is then a continuous $f : X \to \mathbb{R}$ so that $f|_F = \mathbf{0}$ and $f|_{X \setminus U} = \mathbf{1}$. Then $f \in [\mathbf{0}; F, \varepsilon]$.

$\square$

Complete Regularity

Proposition

Suppose $\{f_n : n \in \mathbb{N}\} \in \Omega_{C_p(X),\mathbf{0}}$ and that $U_n \subseteq X$ are open so that $f_n|_{X \setminus U_n} = \mathbf{1}$. Then $\{U_n : n \in \mathbb{N}\} \in \Omega(X)$.

Proof.

Suppose $F \subseteq X$ is finite. Then $[\mathbf{0}; F, 1]$ is open nhood of $\mathbf{0}$, so there is an n so that $f_n \in [\mathbf{0}; F, 1]$. Thus $F \subseteq f_n^{-1}[(-1, 1)]$. Since $f_n|_{X \setminus U_n} = \mathbf{1}$, this means that $F \cap (X \setminus U_n) = \emptyset$. Therefore $F \subseteq U_n$. $\square$

General Translation

Theorem (Caruvana and Holshouser 2019)

Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{D}$ be collections. Suppose there are functions

- $\overleftarrow{T}_{\mathrm{I},n} : \mathcal{B} \to \mathcal{A}$ and
- $\overrightarrow{T}_{\mathrm{II},n} : \bigcup \mathcal{A} \times \mathcal{B} \to \bigcup \mathcal{B}$

so that

1. if $x \in \overleftarrow{T}_{\mathrm{I},n}(B)$, then $\overrightarrow{T}_{\mathrm{II},n}(x, B) \in B$ and
2. if $x_n \in \overleftarrow{T}_{\mathrm{I},n}(B_n)$ for all n, then

$$\{x_n : n \in \mathbb{N}\} \in \mathcal{C} \implies \left\{\overrightarrow{T}_{\mathrm{II},n}(x_n, B_n) : n \in \mathbb{N}\right\} \in \mathcal{D}$$

Then $G_1(\mathcal{A}, \mathcal{C}) \leq_{\mathrm{II}} G_1(\mathcal{B}, \mathcal{D})$.

General Translation

$G_1(\mathcal{A}, \mathcal{E})$: $\qquad \overleftarrow{T}_{I,n} \qquad$ $G_1(\mathcal{B}, \mathcal{D})$:

I	$\overleftarrow{T}_{I,0}(B_0)$	$\overleftarrow{T}_{I,1}(B_1)$	$\cdots$
II	x_0	x_1	$\cdots$

I	B_0	B_1	$\cdots$
II	$\overrightarrow{T}_{II,0}(x_0, B_0)$	$T_{II,1}(x_1, B_1)$	$\cdots$

$\overrightarrow{T}_{II,n}$

Two can win implies Two can win

① $x \in \overleftarrow{T}_{I,n}(B) \Rightarrow \overrightarrow{T}_{II,n}(x, B) \in B$

② $x_n \in \overleftarrow{T}_{I,n}(B_n)$ and $\{x_n : n \in \omega\} \in \mathcal{E} \Rightarrow \{\overrightarrow{T}_{II,n}(x_n, B_n) : n \in \omega\} \in \mathcal{D}$

Using General Translation

Let X be a Hausdorff completely regular space.

Theorem (Caruvana and Holshouser 2019)

$G_1(\Omega_{C_p(X),\mathbf{0}}, \Omega_{C_p(X),\mathbf{0}}) \leq_{\mathrm{II}} G_1(\Omega(X), \Omega(X))$

Define $\overleftarrow{T}_{\mathrm{I},n}(\mathscr{U}) = A(\mathscr{U})$ and set $\overrightarrow{T}_{\mathrm{II},n}(f, \mathscr{U})$ to be a choice of $U \in \mathscr{U}$ so that $f[X \setminus U] = \{1\}$ if possible, and X otherwise. Confirm that

- $\overleftarrow{T}_{\mathrm{I},n}$ and $\overrightarrow{T}_{\mathrm{II},n}$ are well-defined,
- If $f \in \overleftarrow{T}_{\mathrm{I},n}(\mathscr{U})$, then $\overrightarrow{T}_{\mathrm{II},n}(f, \mathscr{U}) \in \mathscr{U}$, and
- If $\mathscr{U}_n$ are ω-covers, f_n and $U_n \in \mathscr{U}_n$ are so that $f_n[X \setminus U_n] = \{1\}$, and $\mathbf{0} \in \overline{\{f_n : n \in \mathbb{N}\}}$, then $\{U_n : n \in \mathbb{N}\}$ is an ω-cover.

The Selection Principle Result

Let X be a Hausdorff completely regular space. Then X is countable if and only if $C_p(X)$ has strong countable fan tightness (this turns out to be equivalent to $C_p(X)$ being metrizable).

A Different Topology

$C_k(X)$

$C_k(X)$ is the topology on $C(X)$ whose open sets are generated by sets of the form

$$[f; K; \varepsilon] = \{g : (\forall x \in K)[|f(x) - g(x)| < \varepsilon]\}$$

where f is continuous, $K \subseteq X$ is compact, and $\varepsilon > 0$

Translation for $C_k(X)$

Caruvana, Holshouser 2019

$G_1(\mathcal{K}(X), \mathcal{K}(X)) \leq_{\mathrm{II}} G_1(\Omega_{C_k(X),\mathbf{0}}, \Omega_{C_k(X),\mathbf{0}})$.

Theorem (Caruvana and Holshouser 2019)

Assume X is a Hausdorff completely regular space. Then $G_1(\Omega_{C_k(X),\mathbf{0}}, \Omega_{C_k(X),\mathbf{0}}) \leq_{\mathrm{II}} G_1(\mathcal{K}(X), \mathcal{K}(X))$.

At the pure selection principle level, this means that X is hemicompact if and only if $C_k(X)$ has strong countable fan tightness (this turns out to be equivalent to $C_k(X)$ being metrizable).

USCO Maps - Definition

USCO Maps

A set valued function $\Phi : X \to \mathcal{P}(Y)$ is upper semicontinuous if for every open $V \subseteq Y$,

$$\Phi^{\leftarrow}[V] = \{x \in X : \Phi(x) \subseteq V\}$$

is open.
Φ is called an **usco** map if it is upper semicontinuous and $\Phi(x)$ is compact for all $x \in X$.
Φ is **minimal** if its graph is minimal with respect to $\subseteq$.

Topologies on USCO Maps

Fix a topological space X. We define the topology on the usco maps from X to $\mathbb{R}$ as a uniform topology.

- First, for $Y \subseteq \mathbb{R}$ and $\varepsilon > 0$, let $\mathbb{B}(Y, \varepsilon) = \bigcup_{y \in Y} B(y, \varepsilon)$,
- Then, for $A \subseteq X$ and $\varepsilon > 0$, let $\mathbf{W}(A, \varepsilon)$ be

$$\{(\Phi, \Psi) : (\forall x \in A)[\Phi(x) \subseteq \mathbb{B}(\Psi(x), \varepsilon) \text{ and } \Psi(x) \subseteq \mathbb{B}(\Phi(x), \varepsilon)]\}$$

- Finally, $[\Phi; A, \varepsilon] = \mathbb{W}(A, \varepsilon)[\Phi]$.
- $\mathrm{MU}_p(X)$ is the topology on the minimal usco maps generated by the $[\Phi; A, \varepsilon]$, where A ranges over the finite subsets of X.
- $\mathrm{MU}_k(X)$ is the topology on minimal usco maps generated by the $[\Phi; A, \varepsilon]$, where A ranges over the compact subsets of X.

Translation for USCO Maps

Thanks to a clever characterization of minimal usco maps by Holá and Holý in 2009, the work on continuous real valued functions can be extended.

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Assume X is a Hausdorff completely regular space. Then $G_1(\Omega_{MU_p(X),\mathbf{0}}, \Omega_{MU_p(X),\mathbf{0}}) \equiv G_1(\Omega(X), \Omega(X))$ and $G_1(\Omega_{MU_k(X),\mathbf{0}}, \Omega_{MU_k(X),\mathbf{0}}) \equiv G_1(\mathcal{K}(X), \mathcal{K}(X))$.

Again, this means that $MU_p(X)$ is metrizable if and only if X is countable.

Thanks!

Thanks for Listening